

- Q.22 Describe boundary layer phenomenon.
 Q.23 What is the use of a supercritical airfoil?
 Q.24 Name the various types of rotorcrafts with their applications?
 Q.25 How do you record speed of an aircraft in flight.
 Q.26 With the help of a sketch show various high lift devices.
 Q.27 Describe stability of an airplane.
 Q.28 Draw various forces on an airplane in climbing attitude.
 Q.29 Write a brief note on helicopter flying.
 Q.30 Describe working of trim tab control.
 Q.31 Suggest methods to reduce induced drag.
 Q.32 List the various regime of flight speed.
 Q.33 Explain flyover concept.
 Q.34 What is adverse pressure gradient?
 Q.35 How does temperature varies in troposphere?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 What is viscosity? How does it create Boundary Layer? What are various methods to control boundary layer?
 Q.37 What is stability? What are various types of it? Explain lateral stability.
 Q.38 Explain working of all the control surfaces used of an airplane.

No. of Printed Pages : 4
 Roll No.

187736/147736

3rd Sem / Aircraft Maintenance Engg. Subject:- Theory of Flight

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following do not INTERSECT each other?
 a) Streakline
 b) Streamline
 c) Path line
 d) Streamline and pathline
 Q.2 Which stability horizontal tail does provide?
 a) Longitudinal b) Lateral
 c) Rolling d) None of the above
 Q.3 What is the primary principle governing aerodynamics?
 a) Newton's First Law of Motion
 b) Bernoulli's Principle
 c) Archimedes's Principle
 d) Ohm's Law

- Q.4 What is the primary means of producing lift in an aircraft?
- Thrust from the engines
 - Weight of the aircraft
 - Pressure difference between the upper and lower surfaces of the wings
 - Control surfaces such as ailerons and elevators
- Q.5 Which control surface is primarily responsible for controlling yaw?
- Elevator
 - Aileron
 - Rudder
 - Flap
- Q.6 What is the thickness of NACA 0002 airfoil?
- 3%
 - 1%
 - 0%
 - 2%
- Q.7 What is the primary function of slats in an aircraft's secondary flying controls?
- To reduce lift and increase drag
 - To increase lift and reduce stall speed
 - To improve roll control
 - To stabilize the aircraft's pitch attitude
- Q.8 Lift to drag ratio is maximum in?
- Airliner
 - Fighter
 - Glider
 - Kite

(2)

187736/147736

- Q.9 The temperature in troposphere
- Decreases with height
 - Increases with height
 - Remains constant
 - Is random
- Q.10 The critical Mach number is
- Between 0 and 1
 - Between 0.7 to 0.9
 - Between 1.0 to 1.2
 - Between 0.8 to 1.0

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What remains constant according to Bernoulli equation?
- Q.12 What is the use of trim tab?
- Q.13 What is the direction of lift on airplane?
- Q.14 What is constant in stratosphere?
- Q.15 What is boundary layer?
- Q.16 How does stall affect aircraft behavior?
- Q.17 What is ISA?
- Q.18 How does tail stabilize airplane?
- Q.19 What is induced Drag?
- Q.20 What are the high lift devices?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe various Aerodynamics forces in aircrafts?

(3)

187736/147736